

Python

main.py

Visualize

Run

```
1 def dice_throw(num_dice, faces, target_sum):
2     ways = [[0] * (target_sum + 1) for _ in range
3             (num_dice + 1)]
4     ways[0][0] = 1
5     for i in range(1, num_dice + 1):
6         for j in range(1, target_sum + 1):
7             total = 0
8             for k in range(1, faces + 1):
9                 if j - k >= 0:
10                    total = total + ways[i - 1][j - k]
11            ways[i][j] = total
12    return ways[num_dice][target_sum]
13
14 num_dice = 3
15 faces = 4
16 target_sum = 7
17
18 result = dice_throw(num_dice, faces, target_sum)
19 print("Number of ways =", result)
20
```

Output

Number of ways = 12

=== Code Execution Successful ===



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Output

```
1
2 def binomial_coefficient(n, r):
3     C = [[0] * (r + 1) for _ in range(n + 1)]
4     for i in range(n + 1):
5         for j in range(min(i, r) + 1):
6             if j == 0 or j == i:
7                 C[i][j] = 1
8             else:
9                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
10    return C[n][r]
11
12
13 n = 5
14 r = 2
15
16 result = binomial_coefficient(n, r)
17 print("C(" + str(n) + "," + str(r) + ") =", result)
```

C(5,2) = 10

=== Code Execution Successful ===



JS

TS





main.py

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```
1 def binomial_coefficient(n, r):
2     C = [[0] * (r + 1) for _ in range(n + 1)]
3     for i in range(n + 1):
4         for j in range(min(i, r) + 1):
5             if j == 0 or j == i:
6                 C[i][j] = 1
7             else:
8                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
9     return C[n][r]
10
11
12 n = 8
13 r = 3
14
15 result = binomial_coefficient(n, r)
16 print("C(" + str(n) + "," + str(r) + ") =", result)
```

Output

C(8,3) = 56

=== Code Execution Successful ===



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```
1 def binomial_coefficient(n, r):
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3     for i in range(n + 1):
4         for j in range(min(i, r) + 1):
5             if j == 0 or j == i:
6                 C[i][j] = 1
7             else:
8                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
9     return C[n][r]
10
11
12 n = 10
13 r = 4
14
15 result = binomial_coefficient(n, r)
16 print("C(" + str(n) + "," + str(r) + ") =", result)
```

Output

C(10,4) = 210

=== Code Execution Successful ===



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```
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3     for i in range(n + 1):
4         for j in range(min(i, r) + 1):
5             if j == 0 or j == i:
6                 C[i][j] = 1
7             else:
8                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
9     return C[n][r]
10
11
12 n = 12
13 r = 5
14
15 result = binomial_coefficient(n, r)
16 print("C(" + str(n) + "," + str(r) + ") =", result)
17
18
```

Output

C(12,5) = 792

=== Code Execution Successful ===



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```
1
2 def binomial_coefficient(n, r):
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4     for i in range(n + 1):
5         for j in range(min(i, r) + 1):
6             if j == 0 or j == i:
7                 C[i][j] = 1
8             else:
9                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
10    return C[n][r]
11
12
13 n = 15
14 r = 6
15
16 result = binomial_coefficient(n, r)
17 print("C(" + str(n) + ", " + str(r) + ") =", result)
18
19
```

Output

C(15,6) = 5005

=== Code Execution Successful ===

https://programiz.pro/?utm_source=compiler



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5             if j == 0 or j == i:
6                 C[i][j] = 1
7             else:
8                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
9     return C[n][r]
10
11
12 n = 10
13 r = 3
14
15 result = binomial_coefficient(n, r)
16 print("C(" + str(n) + "," + str(r) + ") =", result)
```

Output

```
C(10,3) = 120

=== Code Execution Successful ===
```



main.py

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Run

```
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4         for j in range(min(i, r) + 1):
5             if j == 0 or j == i:
6                 C[i][j] = 1
7             else:
8                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
9     return C[n][r]
10
11
12 n = 12
13 r = 4
14
15 result = binomial_coefficient(n, r)
16 print("C(" + str(n) + "," + str(r) + ") =", result)
17
```

Output

```
C(12,4) = 495

=== Code Execution Successful ===
```




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```
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4         for j in range(min(i, r) + 1):
5             if j == 0 or j == i:
6                 C[i][j] = 1
7             else:
8                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
9     return C[n][r]
10
11
12 n = 15
13 r = 5
14
15 result = binomial_coefficient(n, r)
16 print("C(" + str(n) + "," + str(r) + ") =", result)
17
```

Output

C(15,5) = 3003

=== Code Execution Successful ===

<https://adclick.g.doubleclick.net/pcs/click?xai=AKAOjstMM50zY11dAvbFINB2fw1rA...>



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```
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2     C = [[0] * (r + 1) for _ in range(n + 1)]
3     for i in range(n + 1):
4         for j in range(min(i, r) + 1):
5             if j == 0 or j == i:
6                 C[i][j] = 1
7             else:
8                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
9     return C[n][r]
10
11
12 n = 20
13 r = 6
14
15 result = binomial_coefficient(n, r)
16 print("C(" + str(n) + "," + str(r) + ") =", result)
17
```

JS
TS
GO

<https://adclick.g.doubleclick.net/pcs/click?xai=AKAOjstMM50zY11dAvbFINB2fw1rA...>

Output

C(20,6) = 38760

=== Code Execution Successful ===

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```
1
2 def binomial_coefficient(n, r):
3     C = [[0] * (r + 1) for _ in range(n + 1)]
4     for i in range(n + 1):
5         for j in range(min(i, r) + 1):
6             if j == 0 or j == i:
7                 C[i][j] = 1
8             else:
9                 C[i][j] = C[i - 1][j - 1] + C[i - 1][j]
10    return C[n][r]
11
12
13 n = 8
14 r = 3
15
16 result = binomial_coefficient(n, r)
17 print("C(" + str(n) + "," + str(r) + ") =", result)
```

Output

C(8,3) = 56

=== Code Execution Successful ===

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